Qiyang Hong

Ph.D. in Biomedical Engineering

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EDUCATION

- **Ph.D. in Biomedical Engineering**, Peking Union Medical College, Beijing, China (Sept 1, 2023–June 18, 2026)

  Advisor: Prof. Erping Long

  Degree awarded: Doctoral Degree in Engineering (June 26, 2026)

  The degree certificate bears the signatures of the presidents of **Peking Union Medical College** and **Tsinghua University**.

  Doctoral research: Developed ukbFound using UK Biobank data to identify patient subgroups, study how diseases co-occur, and predict disease risk (npj Digital Medicine, 2026).

  Thesis defense: May 25, 2026
- **M.S. in Biochemistry and Molecular Biology**, Xiamen University, Xiamen, China (Sept 13, 2014–June 30, 2017)

  Advisor: Prof. Ning-Shao Xia

  Research contribution: Bioinformatics support, data processing, and computational analysis.
- **B.Eng. in Bioengineering**, Jimei University, Xiamen, China (Sept 1, 2010–July 1, 2014)
- **B.Eng. in Computer Science and Technology,** The Open University of China, Beijing, China (Sept 1, 2023–Jan 20, 2026)

PUBLICATIONS

Published Articles

** First-listed author; † Co-first authors (equal contribution).*

1. Yi E†, Cui J†, Wang H†, Wu F†, **Hong Q**†, Li Q, Xie C, Xu H, Liu Y, Ran X, Wu X, Wan Q, Tang G, Zhu L, Pang J, Zhou Y, Long E, Ran P. Multiomics Mendelian randomization identifies serpin family G member 1 as a chronic obstructive pulmonary disease modulator. *Signal Transduction and Targeted Therapy.* 2026;11:34. doi:10.1038/s41392-025-02547-7.
2. **Hong Q**†, Wang C†, Wu W, Li Y, Sun X, Huang Z, Wang J, Zhang M, Wan P, Ni X, Yi E, Pang J, Long E. A foundational model encodes deep phenotyping data and enables diverse downstream applications. *npj Digital Medicine*. 2026;9:576. doi:10.1038/s41746-026-02736-w.

3. Sun X†, **Hong Q†**, Zhang M, Li Y, Chen T, Huang Z, Liang G, Tang W, Xu S, Ni X, Pang J, Wan P, Long E. Model confrontation and collaboration: A debate intelligence framework for enhancing medical reasoning in large language models. *Cell Reports Medicine*. 2026;7(1):102547. doi:10.1016/j.xcrm.2025.102547.

4. Wu W†, Zhou M†, Chen J, Wang J, **Hong Q**, Li Y, Wang Y, Zhang P, Huang Y, Zhang Z, Zhang A, Liu S, Hu H, Yin Z, Han Z, Chen Q, Xue Y, Huang K, Zhang M, Yi E, Zhao H, Wang C, Pang J, Wang J, Long E. Deep multi-omics profiling reveals three molecular subtypes of chronic obstructive pulmonary disease in a unique biomass-exposed Chinese population. *Med*. 2026;7(4):101033. doi:10.1016/j.medj.2026.101033.

5. Pang J†, Ren X†, **Hong Q†**, Huang K, Wang J, Li D, Huang X, He R, Jiang J, Li X, Li Y, Wang Z, Xu Y, Xie S, Liang C, Long E, Wang J, Yang T, Wang C. Proteo-metabolomic insights into the progression of chronic obstructive pulmonary disease and lung function decline. *Respiratory Research*. 2026;27:379. doi:10.1186/s12931-026-03768-2.

6. Long Y†, Ni X†, Chen T†, Huang X, **Hong Q**, Wang J, Wang C, Huang Z, Xu H, Sun M, Pang J, Choi J, Zhang T, Long E. Genetic determinants of gene expression noise and its role in complex trait variation. *Cell Reports*. 2025;44(12):116612. doi:10.1016/j.celrep.2025.116612.

7. Lin K†, **Hong Q†**, Fu Y†, Tu H, Lin H, Huang J, Hu Y, Huang M, Chen M. Cervical HPV infection and related diseases among 149,559 women in Fujian: an epidemiological study from 2018 to 2023. *Frontiers in Microbiology*. 2024;15:1418218. doi:10.3389/fmicb.2024.1418218.

8. Huang Y†, Li Z†, **Hong Q**, Zhou L, Ma Y, Hu Y, Xin J, Li T, Kong Z, Zheng Q, Chen Y, Zhao Q, Gu Y, Zhang J, Wang Y, Yu H, Li S, Xia N. A stepwise docking molecular dynamics approach for simulating antibody recognition with substantial conformational changes. *Computational and Structural Biotechnology Journal*. 2022;20:710-720. doi:10.1016/j.csbj.2022.01.012.

9. Xin L*, Yu H, **Hong Q**, Bi X, Zhang X, Zhang Z, Kong Z, Zheng Q, Gu Y, Zhao Q, Zhang J, Li S, Xia N. Identification of strategic residues at the interface of antigen-antibody interactions by in silico mutagenesis. *Interdisciplinary Sciences: Computational Life Sciences*. 2018;10(2):438-448. doi:10.1007/s12539-017-0242-7.

10. Xu L†, Zheng Q†, Li S†, He M, Wu Y, Li Y, Zhu R, Yu H, **Hong Q**, Jiang J, Li Z, Li S, Zhao H, Yang L, Hou W, Wang W, Ye X, Zhang J, Baker TS, Cheng T, Zhou ZH, Yan X, Xia N. Atomic structures of Coxsackievirus A6 and its complex with a neutralizing antibody. *Nature Communications*. 2017;8:505. doi:10.1038/s41467-017-00477-9.

11. Li Z†, Yan X†, Yu H†, Wang D, Song S, Li Y, He M, **Hong Q**, Zheng Q, Zhao Q, Gu Y, Zhang J, Janssen MEW, Cardone G, Olson NH, Baker TS, Li S, Xia N. The C-terminal arm of the human papillomavirus major capsid protein is immunogenic and involved in virus-host interaction. *Structure*. 2016;24(6):874-885. doi:10.1016/j.str.2016.04.008.

12. Zhang X[†], Wang K[†], Lin Q[†], Zheng M, Li Q, Li T, **Hong Q**, Zheng Q, Yu H, Gu Y, Li S, Xia N. A shared N-terminal hydrophobic tail for the formation of nanoparticulates. *Nanomedicine*. 2016;11(17):2289-2303. doi:10.2217/nnm-2016-0146.

Accepted Articles

13. Xie W[*], et al. Integrating Large Language Models Into Patient Health Education: Interview Study Among Nurses. *Journal of Medical Internet Research*. Accepted for publication, October 2026. Manuscript 102846.
doi:10.2196/102846.

Article Accepted in Principle

14. Pang J[†], Wang J[†], Wu S[†], **Hong Q**, Wu W, Long Y, Zhou Y, Zhang Z, Zheng C, Zhang Z, Ren X, Zhai Z, Yang T, Huang K, Long E, Wang J, Wang C. GWAS meta-analyses with a new Chinese cohort expands COPD genetics and enhances ancestry-specific risk prediction. *Genomics, Proteomics & Bioinformatics*. Accepted in principle, 2026.

Published Articles in Chinese

15. **Hong Q[*]**, Bi X, Wang D, Li Z, Yu H, Xia N, Li S. Research Progress on RNA-Seq Technology. *Chinese Journal of Biochemical and Pharmaceuticals*. 2017;37(6):443-448. (In Chinese)
16. Bi X[*], **Hong Q**, Yu H, Li S, Xia N. Progress and Applications of In Silico Prediction of Therapeutic Antibody Aggregation Propensity. *Chinese Journal of Pharmaceuticals*. 2017;48(6). (In Chinese)

TECHNICAL SKILLS

- **Biomedical Data Science:** Analyze clinical and cohort data, model changes in health over time, and evaluate and interpret models for disease risk prediction.
- **Bioinformatics & Statistical Analysis:** Multi-omics, genomic and transcriptomic data analysis; survival analysis, multiple-testing correction, and resampling methods.
- **Programming & Data Analysis:** Python, PyTorch, NumPy, pandas, scikit-learn, and reproducible data-analysis workflows.

PROFESSIONAL EXPERIENCE

- **Group Leader, Molecular Diagnostics (technical staff, not a physician)**
  Affiliated Hospital of Putian University, Prenatal Diagnosis Center
  Aug 18, 2020 – June 5, 2023 | Supervisor: Kun Lin
  Led bioinformatics workflows for clinical genetic testing, including sequencing data analysis, quality control, and variant interpretation. During the COVID-19 pandemic, I also assisted with nucleic acid extraction as assigned by the hospital.
- **Bioinformatics Supervisor**
  Berry Genomics Co., Ltd. | Supervisor: Fuli Yu, Ph.D.
  Jul 2017 – Aug 2020
  Developed and maintained automated bioinformatics workflows for genetic testing, including sequencing data analysis and variant annotation. Built web-based tools to support genetic data interpretation and reporting.

TEACHING EXPERIENCE

- **Part-time Course Tutor**

  Experimental College, The Open University of China

  Sept 1, 2023 – Aug 31, 2025

  Provided online part-time tutoring for the course Special Topics in Artificial Intelligence.

PEER REVIEW EXPERIENCE

Independent peer reviewer (2026): Submitted 14 review reports on 13 manuscripts for seven journals, including the Journal of Medical Internet Research, JMIR AI, and JMIR Medical Informatics. Two further reviews are in progress as of November 4, 2026.

Co-reviewer (with Prof. Erping Long): Nature Medicine; Nature Biomedical Engineering; Frontiers in Aging Neuroscience.

RECENT RESEARCH PROJECTS

- **ukbFound for patient stratification and disease risk prediction**

  Independently developed, built, and evaluated ukbFound, a UK Biobank-based model using deep phenotyping data from more than 500,000 participants. The work supported disease risk prediction, multimorbidity analysis, and patient stratification across 289 conditions (npj Digital Medicine, 2026).

- **Evaluation of medical language models**

  Contributed to data acquisition, curation, and analysis for evaluating collaborative language models on medical questions and clinical reasoning tasks (Cell Reports Medicine, 2026).

- **Multi-omics analysis of COPD and lung function decline**

  Contributed to studies integrating genetic, protein, metabolite, and clinical data to investigate chronic obstructive pulmonary disease (COPD), molecular subtypes, and longitudinal lung function decline (Signal Transduction and Targeted Therapy, Med, and Respiratory Research, 2026).